

1. Project Category & Title

- **Category:** Campus Health (Healthcare)
- **Title:** Campus Health Appointment System

2. Problem & Users

- **Problem Statement:** What problem are you solving? (1 paragraph)

Many university students need convenient access to healthcare services, but scheduling appointments and receiving timely communication can be inefficient when handled through phone calls, emails, or manual processes. The Campus Health Appointment System provides a secure online platform where students can schedule appointments, receive confirmation notifications, and manage their healthcare interactions with the student's health center. Healthcare providers can access authorized visit records while role-based access controls help protect sensitive medical information. The system also supports optional communication with a student's primary care provider when authorized by the patient, helping improve coordination of care while respecting privacy requirements.

- **Target Users:**
 - University students seeking appointments
 - University employed doctors and healthcare staff
 - Administrative staff who assist with scheduling

3. Core Features

- **Must-Have Features:**
 1. User authentication and role-based access control (Doctor and Patient roles)
 2. View appointment slots
 3. Book appointments and create visit records
 4. Send appointment confirmation SMS notifications using Twilio
 5. AI-assisted appointment recommendation that ranks available appointment options based on the student's selected service category, scheduling preferences, and provider availability.
 6. Provide a brief explanation of recommendation factors, such as service match, availability, and scheduling preference.
 7. Allow the student to ignore the AI recommendation and manually select any eligible appointment, preserving human control over the final decision.

8. Validate AI recommendations against actual appointment availability before presenting them to the student and provide a standard scheduling fallback if the AI component is unavailable.

- **Nice-to-Have Features:**

1. Primary Care Provider (PCP) notification request with patient authorization
2. Functionality to cancel or reschedule appointments

4. Architecture & Tech Stack

- **Architecture Sketch:** A simple description or diagram of how it works.

The Campus Health Appointment System is a healthcare scheduling platform designed for university students and campus medical staff. Students can view available appointment slots, book appointments, and receive SMS confirmations through Twilio. Doctors can securely access authorized visit records, and users authenticate through Okta with role-based access controls to protect sensitive healthcare information.

- **Tech Stack:** List the specific tools you plan to use.
 - *Languages:* Python, JavaScript, HTML, CSS
 - *Frameworks:* Flask, React
 - *Data:* SQLite

Repository link for capstone project: <https://github.com/psu-edu/sweng861-capstone-lmg6725>